The Hong Kong Polytechnic University
Department of Building Environment and Energy Engineering

BSE3124 Electricity and Buildings **2026/27 Semester 1**

Assignment – Answer Sheets

Group number: _____

Names of group members: _____

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Task 1 – Loading estimation and transformer room location

Estimation of the building electrical load using CLP COP215 and justifications for transformer room and main LV switch room locations:

Task 2 – Layout drawings and MCB board schematics

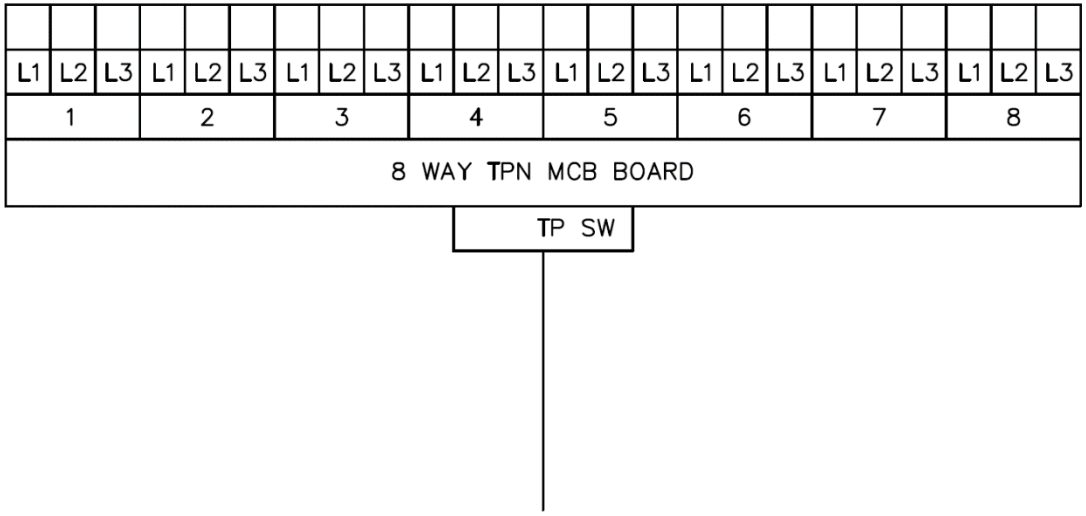
The office tenant chosen for layout design: Office 1 / Office 2 / Office 3

Design considerations of circuit arrangement and protective device selection for small power and lighting layout:

Task 3 – Maximum current demand calculation and cable sizing

Maximum current demand calculation:

Cable sizing for the incoming circuit to the distribution board:



MCB BOARD SCHEMATIC

Notes

Client

Project

Drawing

Date

Project number

Drawing number

Scale